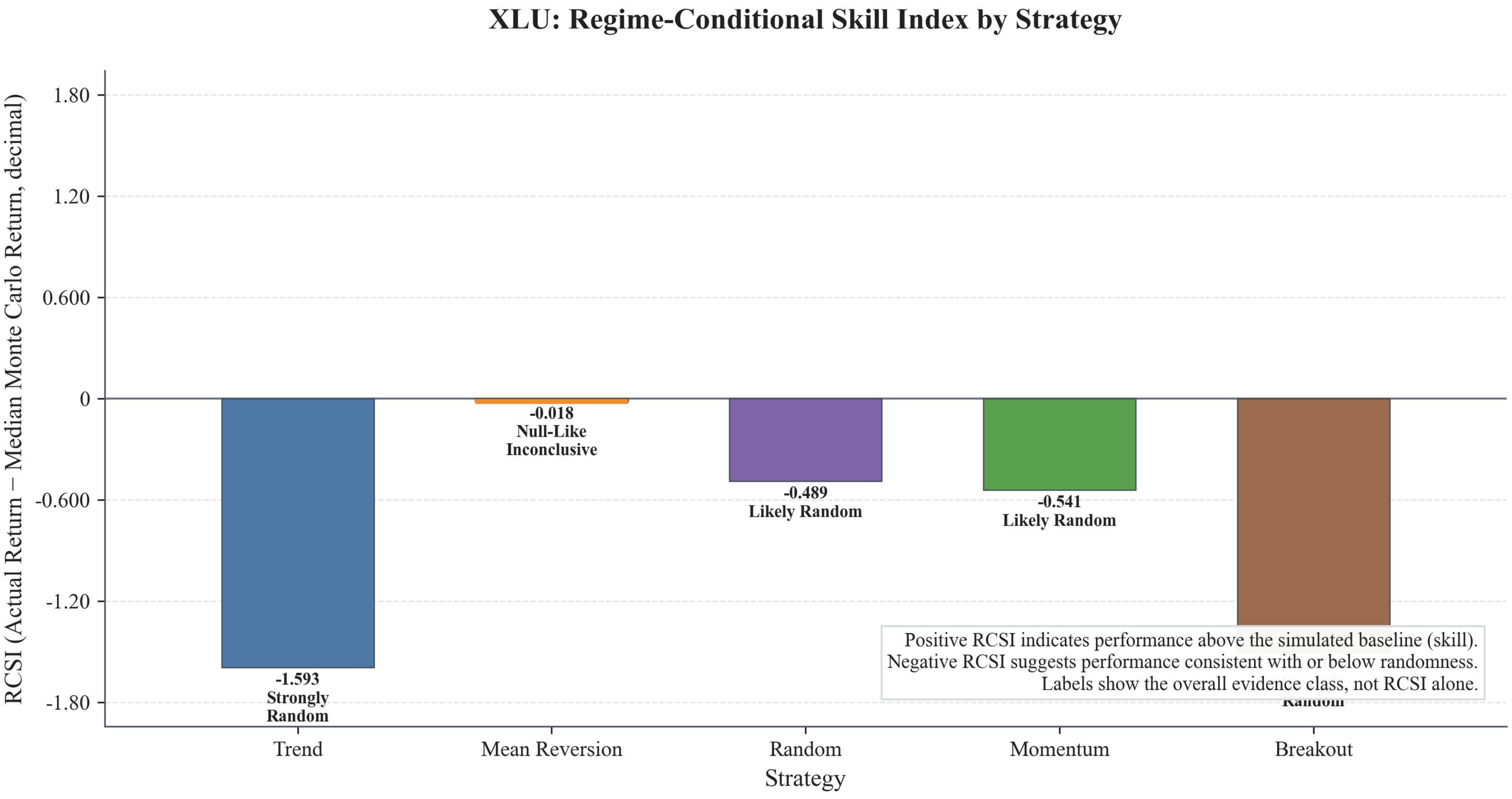
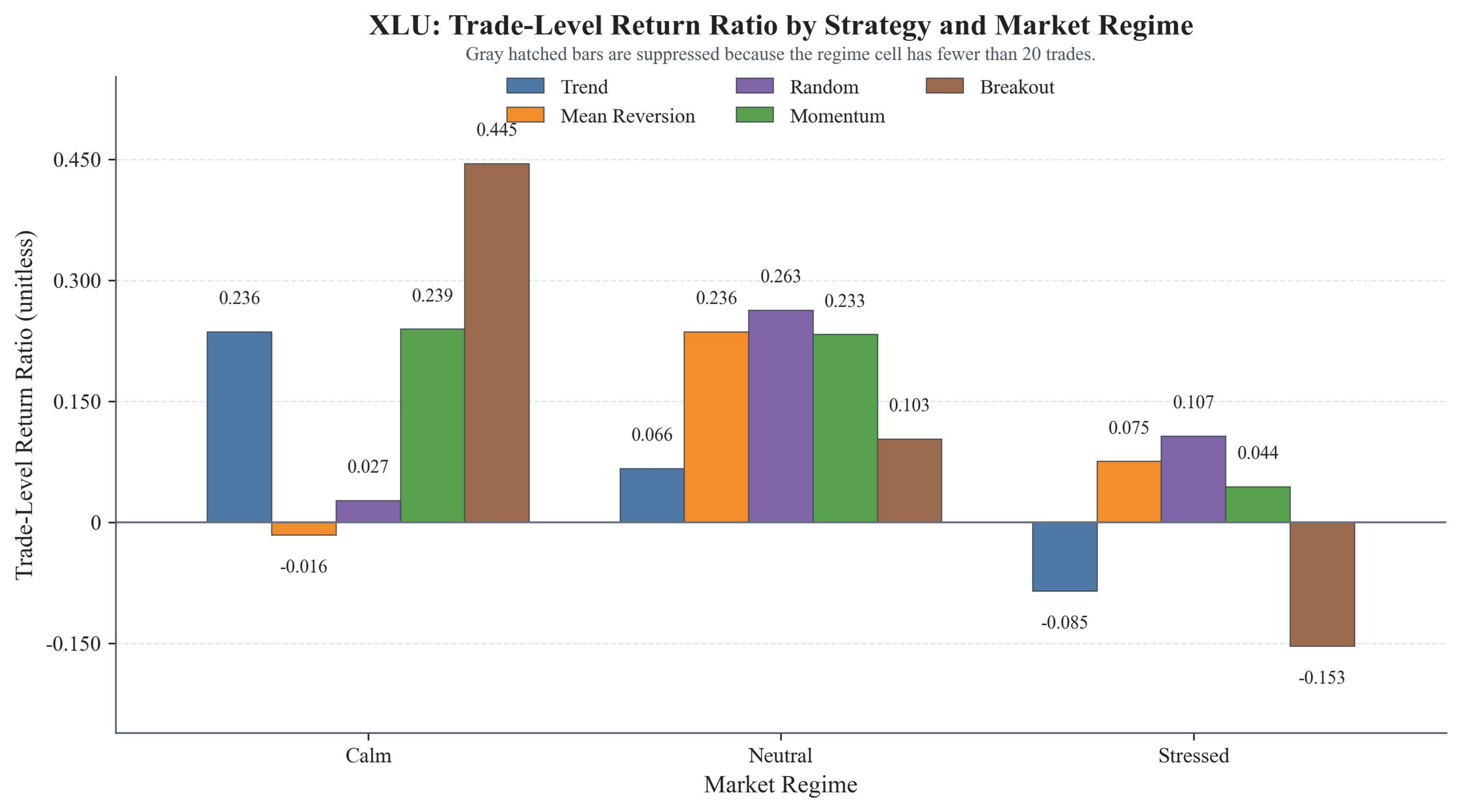
XLU: Strategy Evidence and Confidence Overview

Evidence labels use scale-free statistics: p-value, percentile, normalized RCSI, robustness, and stability.

Strategy	Evidence	Verdict	Confidence	p-value	Prom.	Pctile	RCSI_z
Trend	Strongly Random	Luck	High	0.945	0.02	5.5	-1.20
Mean Reversion	Null-Like / Inconclusive	Inconclusive	Low	0.510	0.29	49.0	-0.17
Random	Likely Random	Luck	Moderate	0.706	0.15	29.4	-0.62
Momentum	Likely Random	Luck	Moderate	0.653	0.18	34.7	-0.51
Breakout	Strongly Random	Luck	High	0.911	0.04	8.9	-1.07

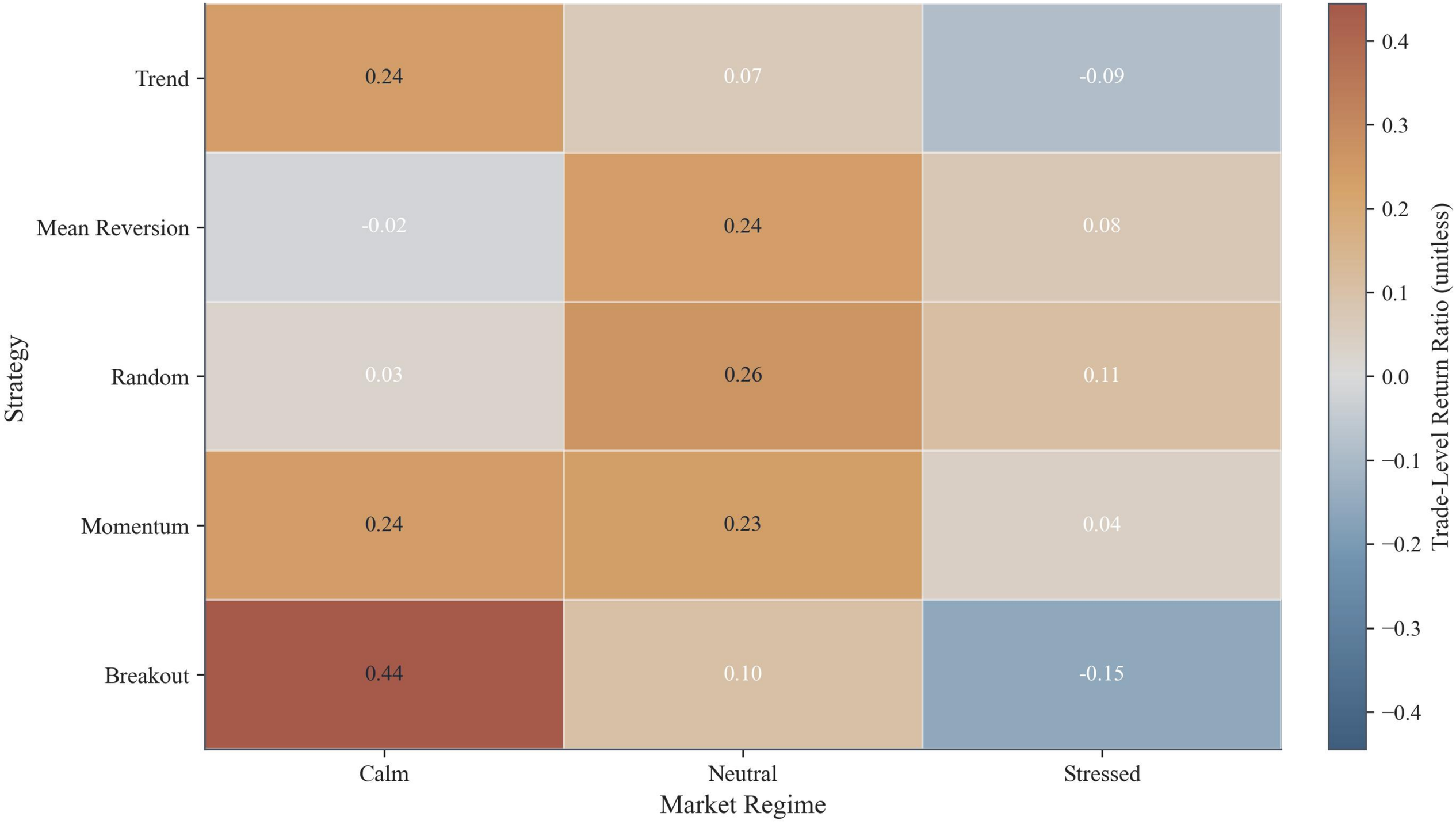
Interpretation:
Prom. = -log10(p), so larger values indicate rarer outcomes under the null.
Pctile and RCSI_z provide scale-free evidence and are more comparable across tickers.
Benchmark context appears in the comparison table and equity curve rather than this verdict summary.


XLU: Regime-Conditional Skill Index by Strategy
RCSI (Actual Return − Median Monte Carlo Return, decimal)
1.80
1.20
0.600
0
-0.600
-1.20
-1.80
-1.593
Strongly
Random
-0.018
Null-Like
Inconclusive
-0.489
Likely Random
-0.541
Likely Random
Positive RCSI indicates performance above the simulated baseline (skill).
Negative RCSI suggests performance consistent with or below randomness.
Labels show the overall evidence class, not RCSI alone.
Trend
Mean Reversion
Random
Momentum
Breakout
Strategy

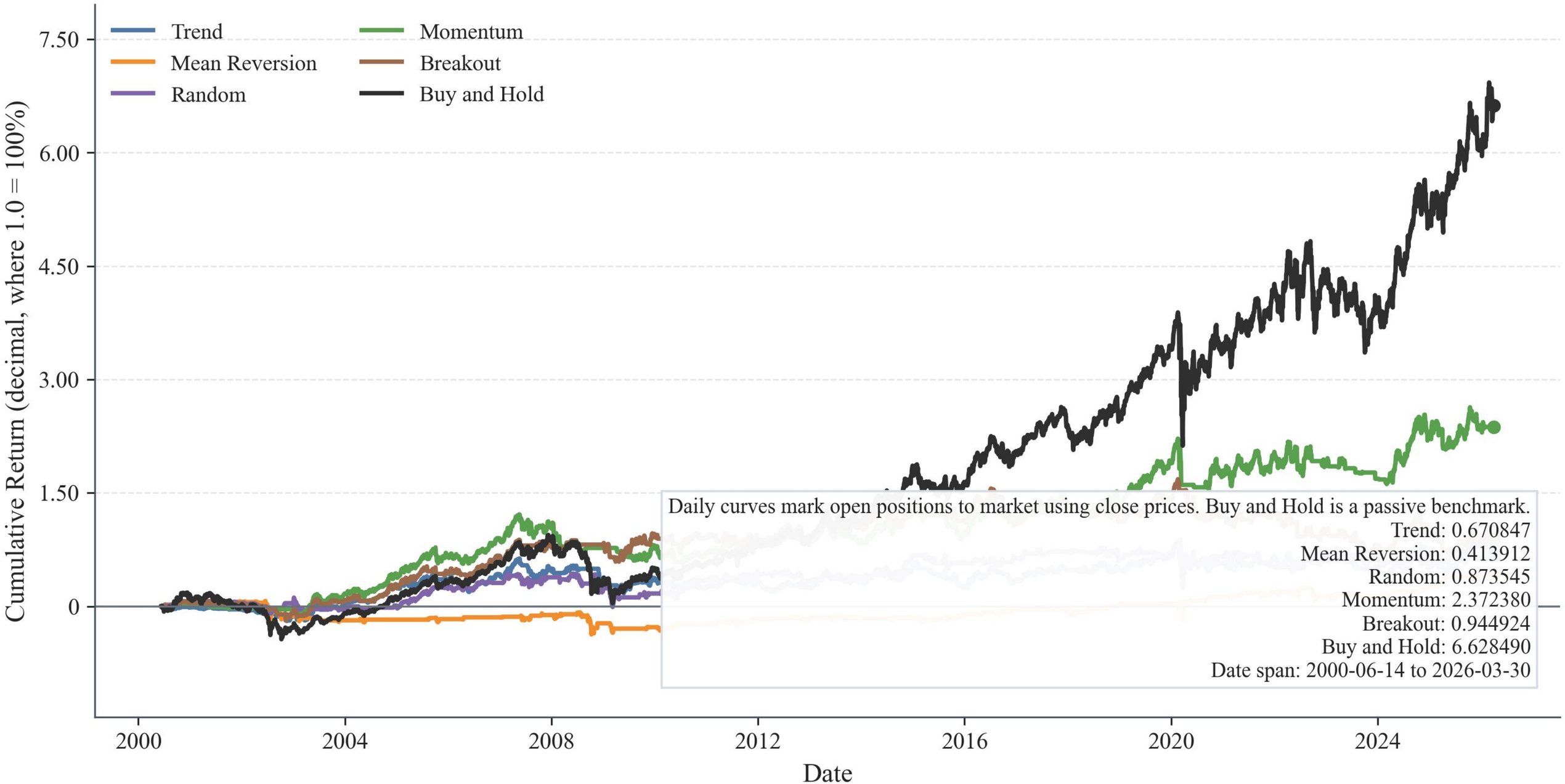

XLU: Trade-Level Return Ratio by Strategy and Market Regime
Gray hatched bars are suppressed because the regime cell has fewer than 20 trades.
Trend
Mean Reversion
Random
Momentum
Breakout
Trade-Level Return Ratio (unitless)
0.450
0.300
0.150
0
-0.150
0.236
-0.016
0.027
0.239
0.445
0.066
0.236
0.263
0.233
0.103
-0.085
0.075
0.107
0.044
-0.153
Calm
Neutral
Stressed
Market Regime

XLU: Strategy Performance Across Market Regimes

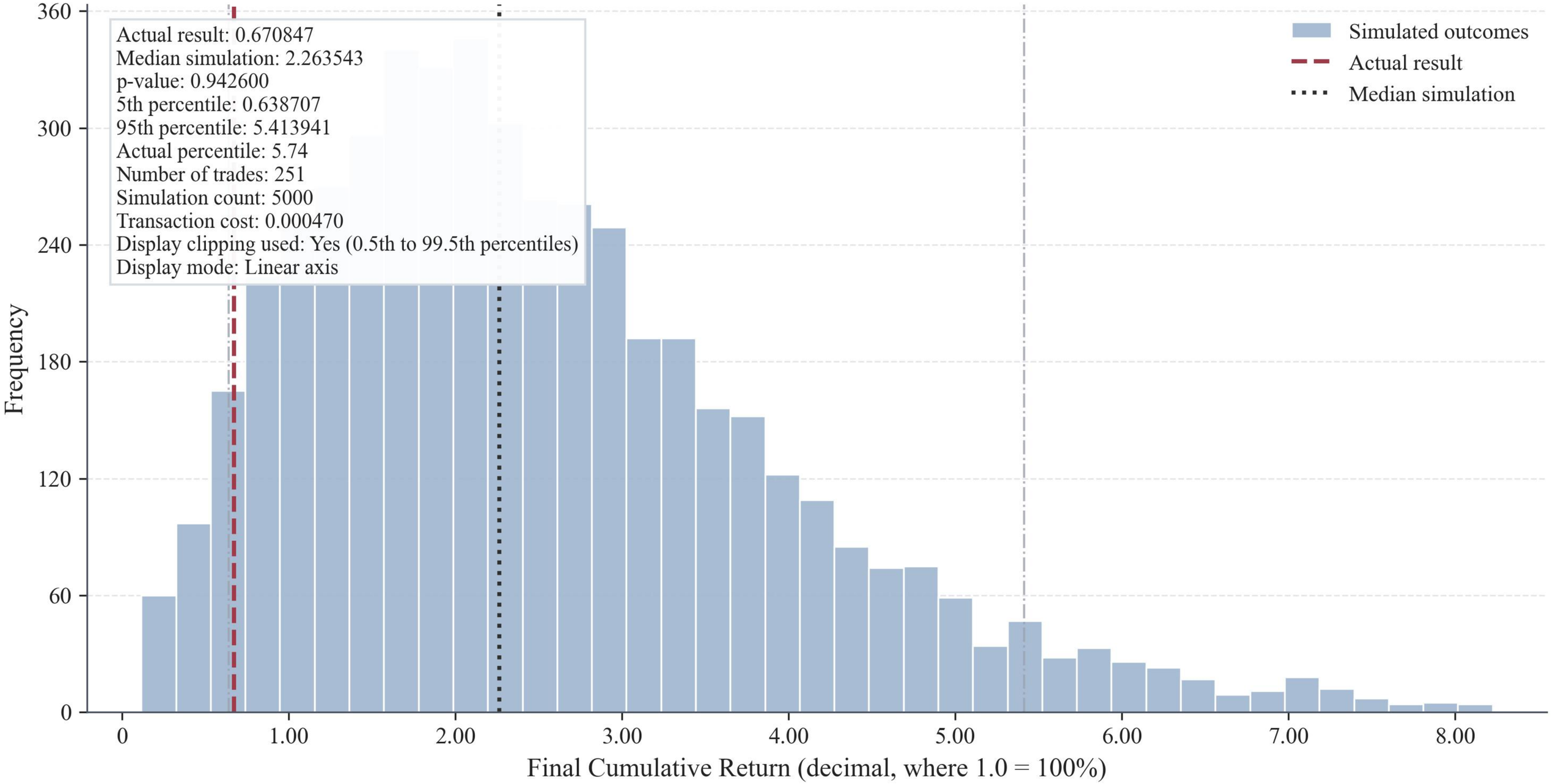
Gray cells are suppressed because the regime cell has fewer than 20 trades.



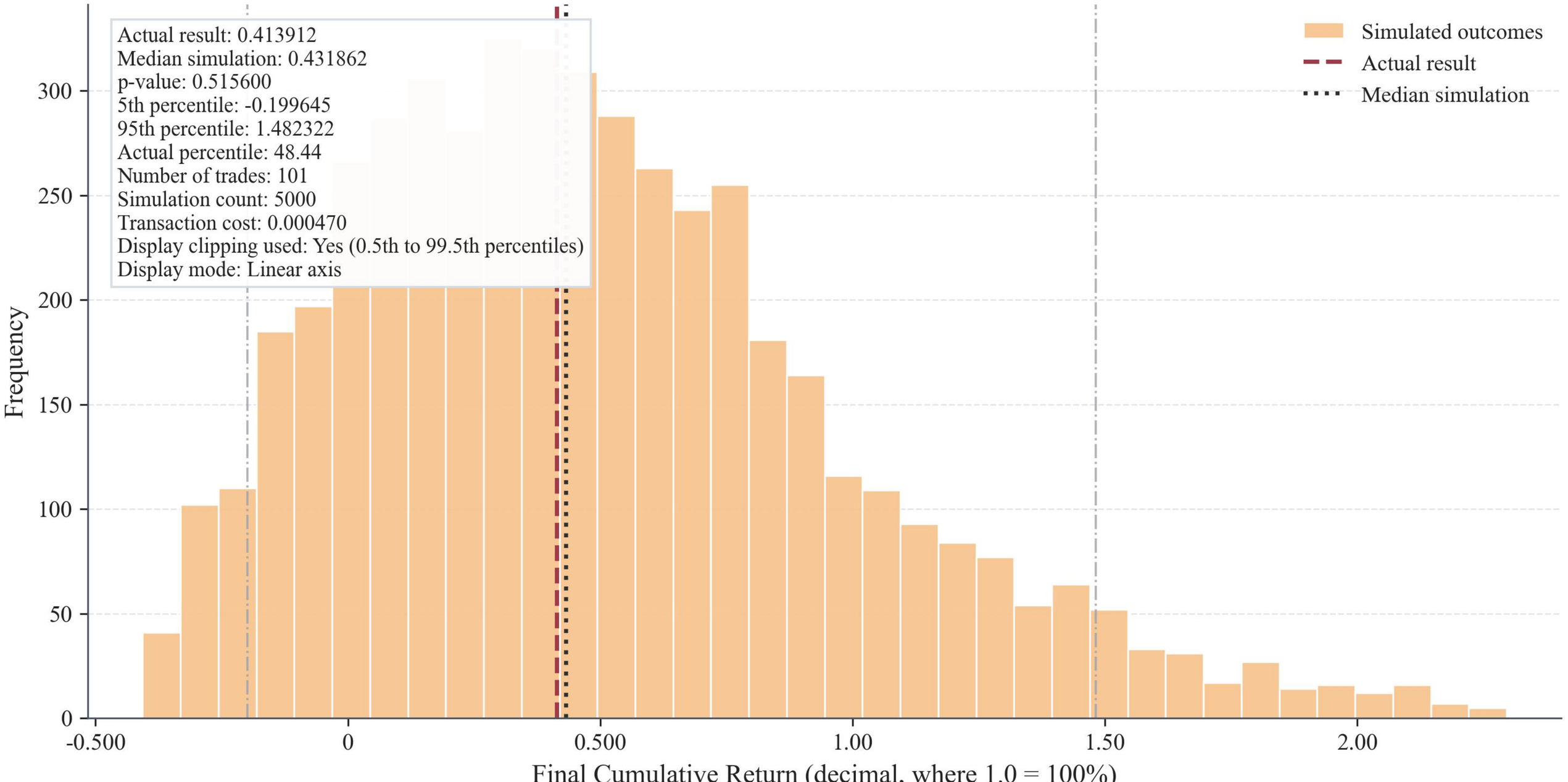
XLU: Daily Equity Curves by Strategy and Benchmark



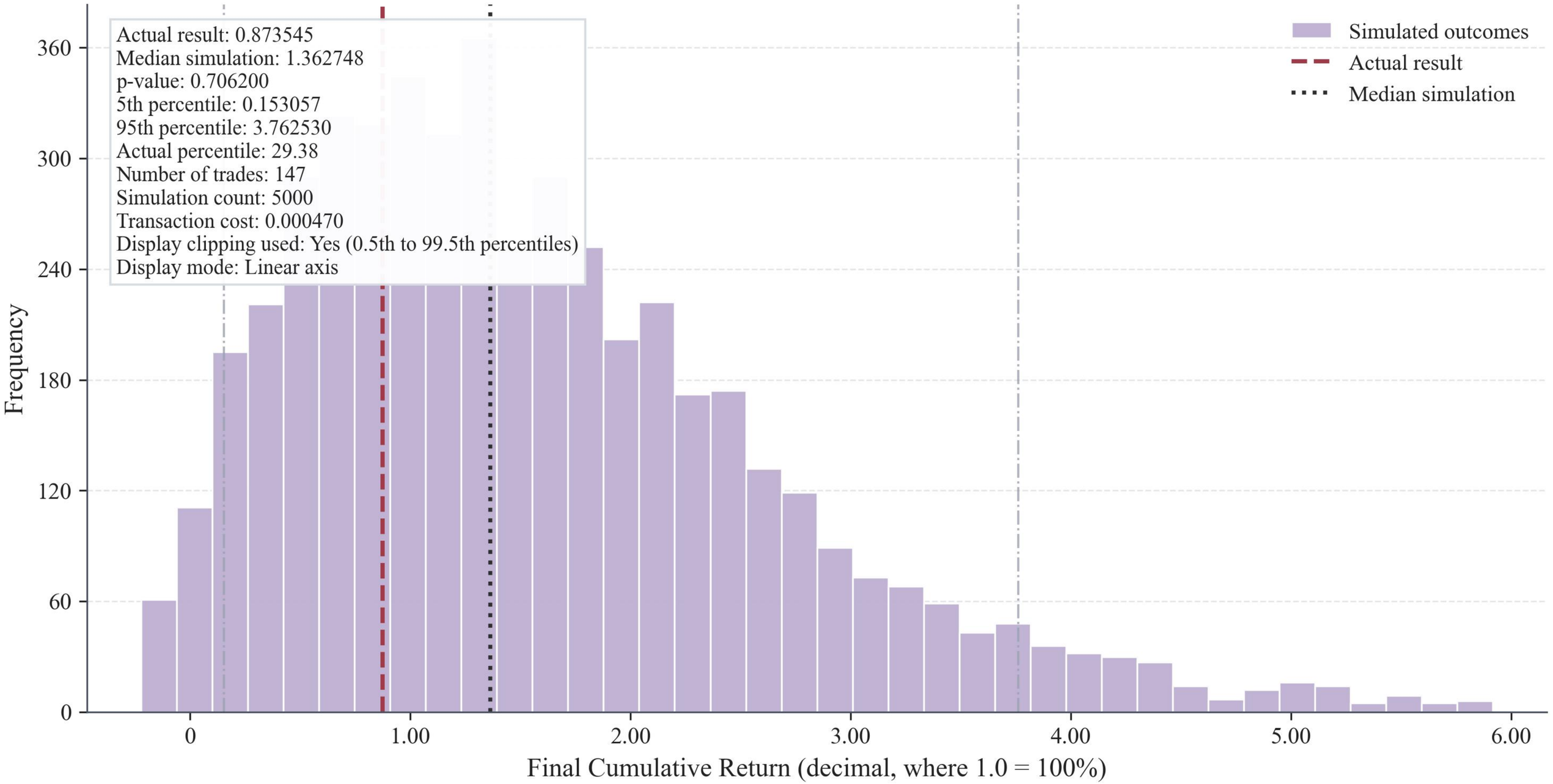
XLU: Monte Carlo Distribution of Final Returns, Trend Strategy



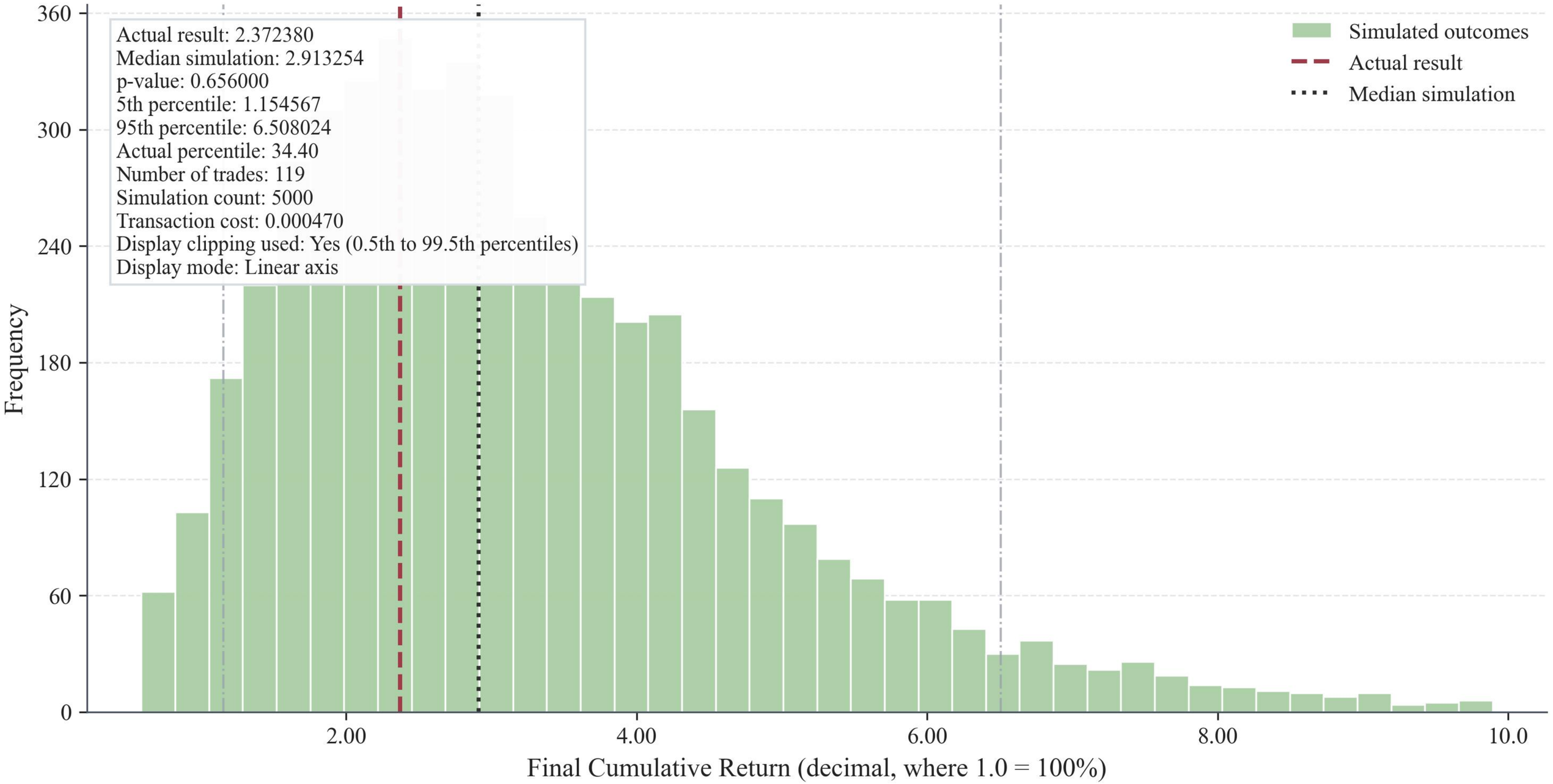
XLU: Monte Carlo Distribution of Final Returns, Mean Reversion Strategy



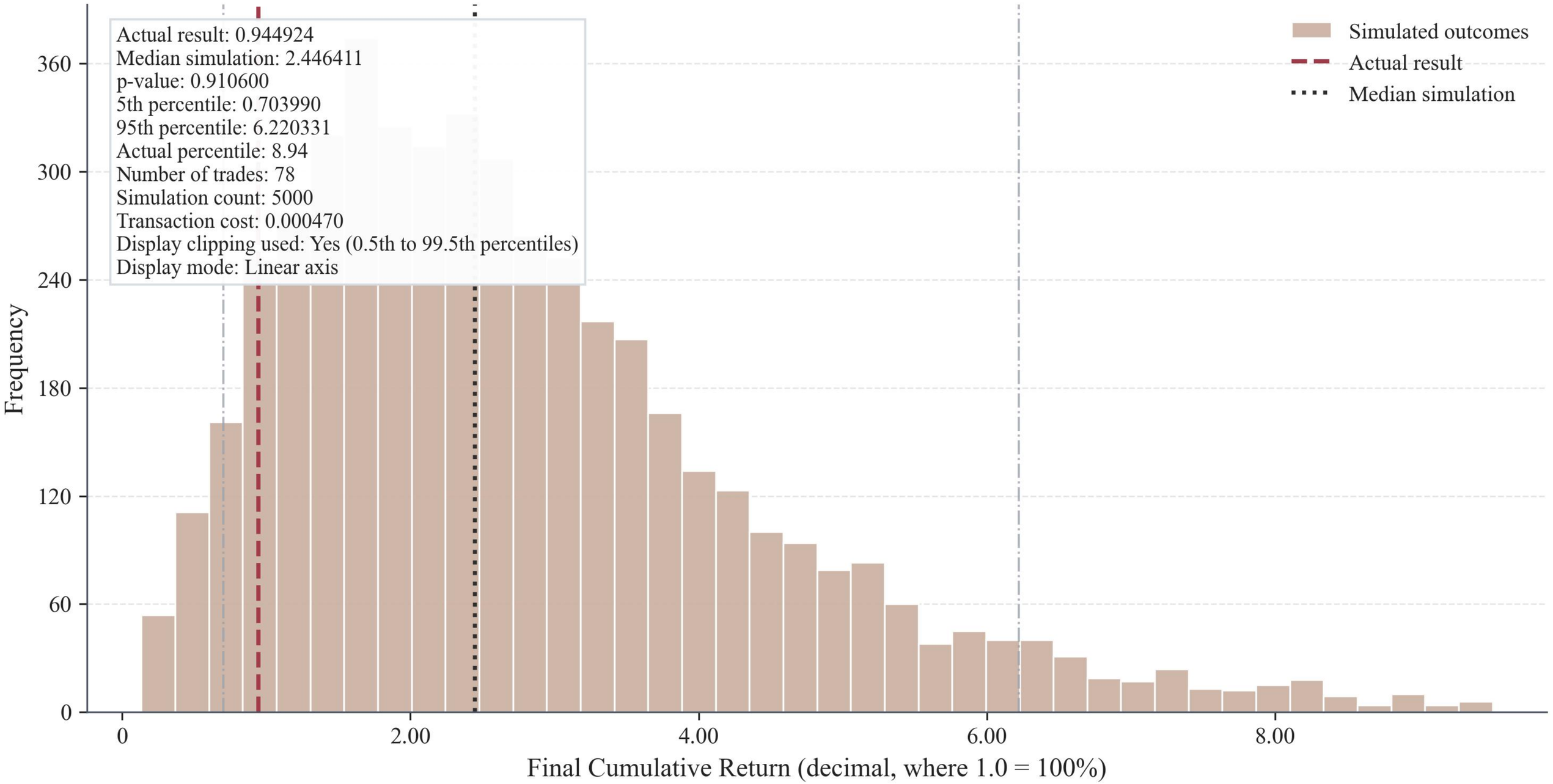
XLU: Monte Carlo Distribution of Final Returns, Random Strategy



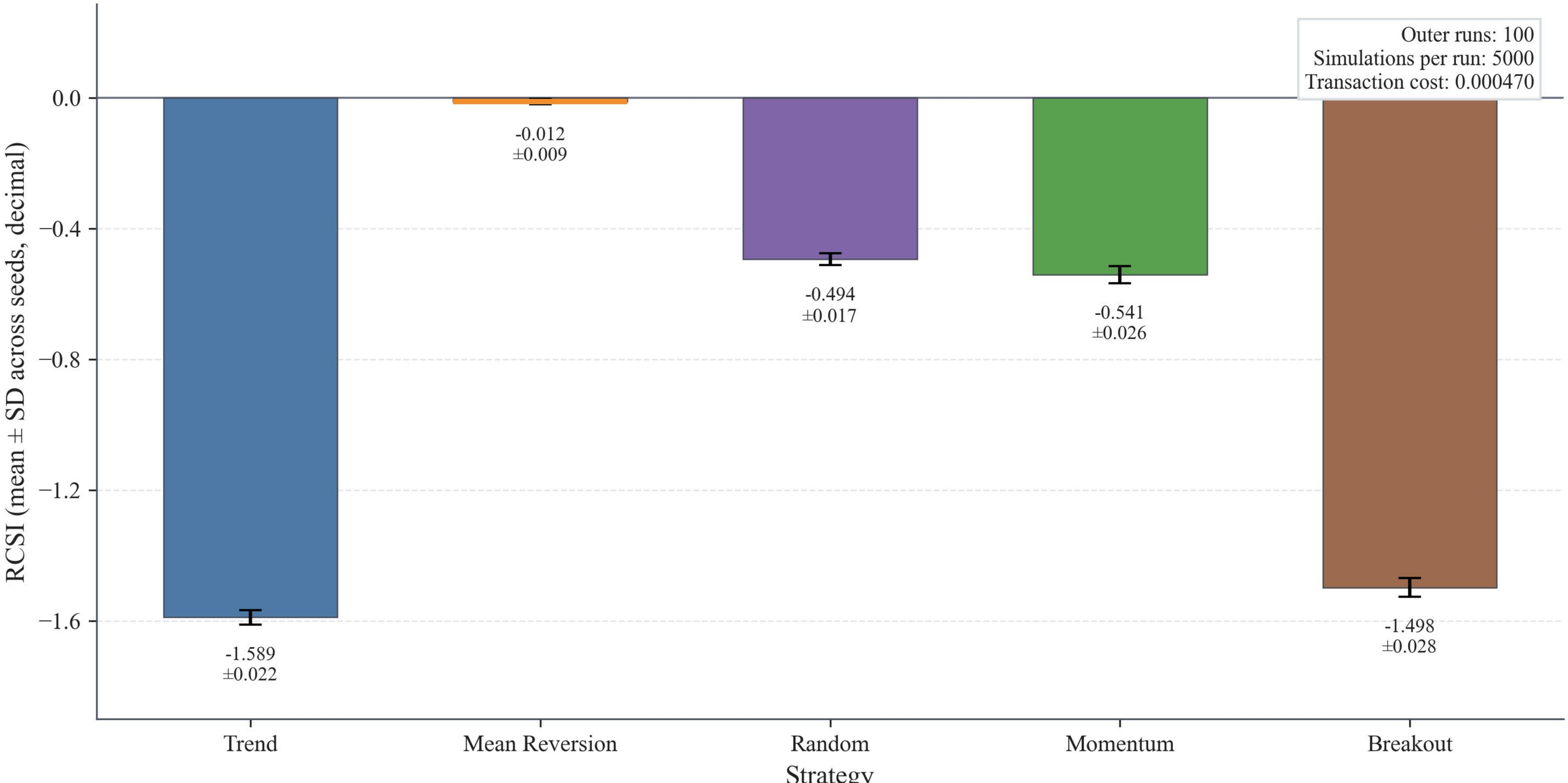
XLU: Monte Carlo Distribution of Final Returns, Momentum Strategy



XLU: Monte Carlo Distribution of Final Returns, Breakout Strategy

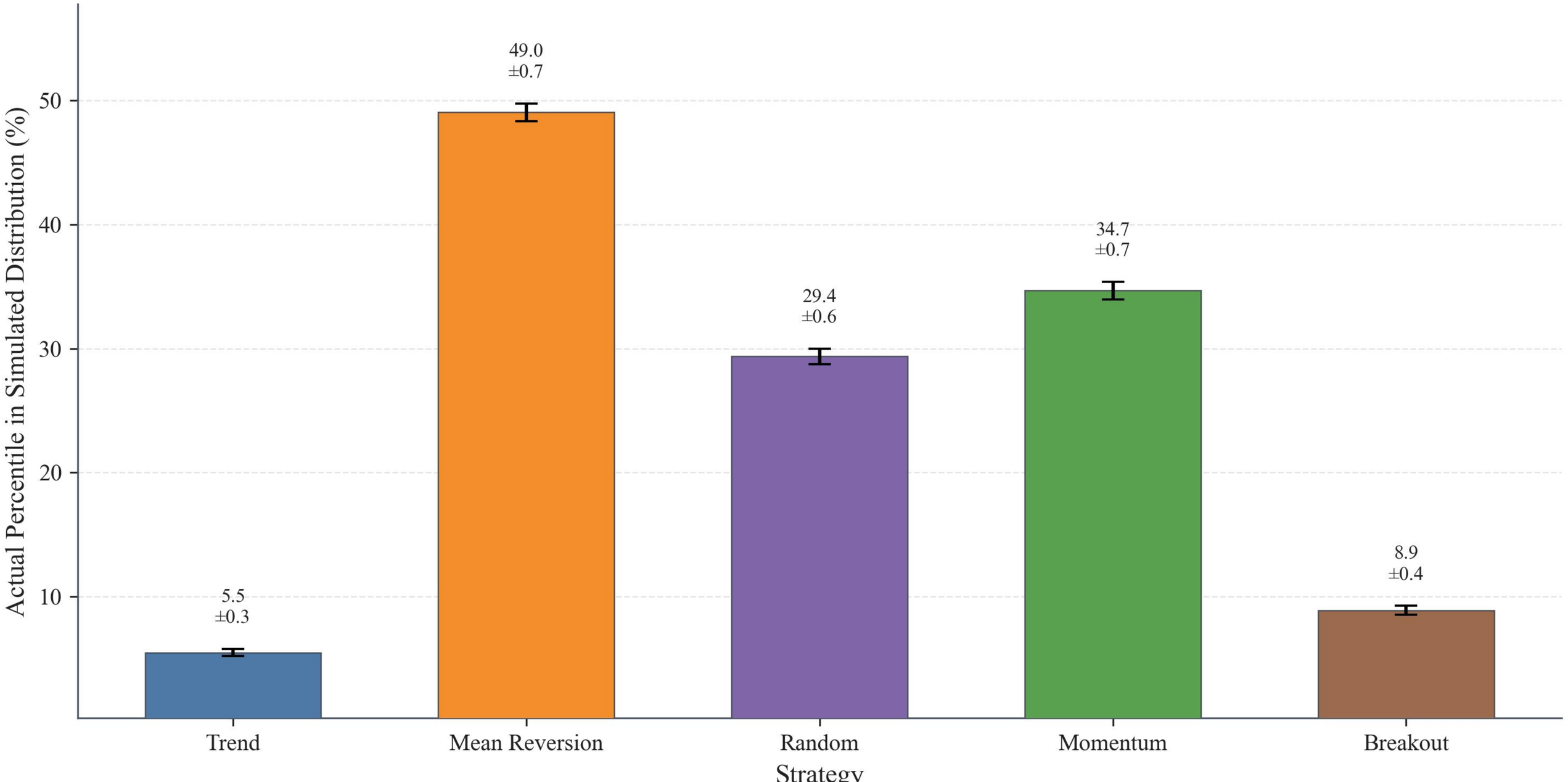


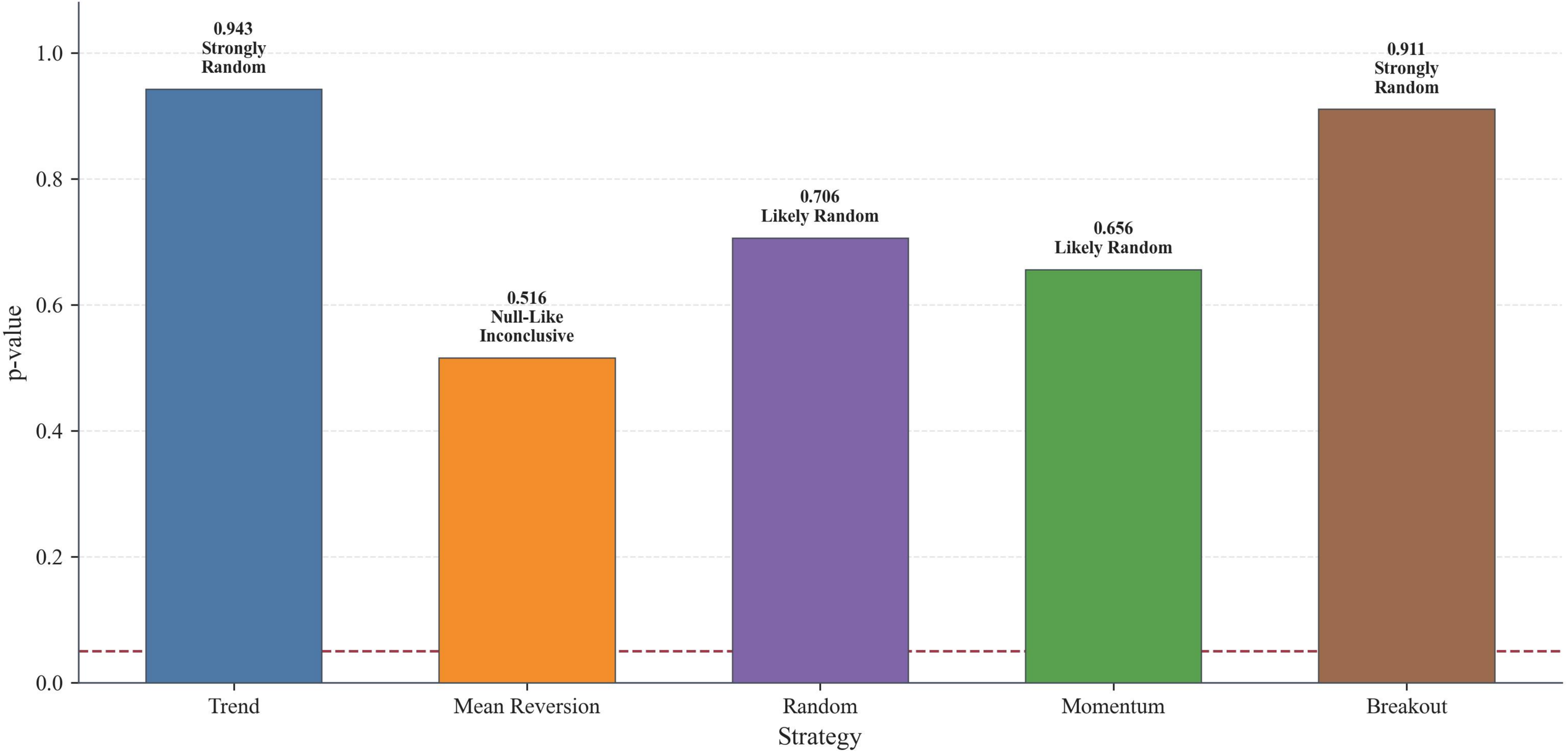
XLU: Monte Carlo Robustness of RCSI Across Seeds



XLU: Monte Carlo Robustness of Actual Percentile Across Seeds

Higher values place the observed outcome deeper into the right tail of the simulated distribution.



XLU: Monte Carlo p-Value by Strategy

The dashed line marks p = 0.05. Lower values indicate stronger evidence against the simulated-random baseline. Labels show the overall evidence class.